Routing Is Least Learnable Where It Is Most Valuable: Bounds on Representation Routing for Web Agents

Anonymous ACL submission

Abstract

A web agent can read a page as text, look at a screenshot, or do both, and deployments usually fix one choice for every task. We measure that choice across three sites of VisualWebArena and WebArena with four vision-language backbones, using screenshot-free variants as controls and rerunning identical conditions to see which differences survive. Three findings separate questions that are usually merged. First, no observation choice is best everywhere. The screenshot's value is real, reproducible and, on one site, predictable in advance. A zero-cost rule over the task text flags classifieds tasks where adding the screenshot alone raises success by 25–30 points for the two strongest backbones (11 for a 4B model). For the 235B backbone the gap reappears on a full rerun. Second, per-task preferences between representations are reproducible beyond an additive model of task difficulty. Yet single-run labels measure them with reliability 0.31–0.50 and, over six modes, flip most where the choice can change success, as that model predicts. Third, two lightweight routers choose a representation per task from pre-run features. With their operating point fixed on training data, they show no reliable gain over the fixed choices and their random mixtures. We describe the rerun-controlled protocol that keeps these three questions apart.

1 Introduction

Web agents can perceive a page in two standard ways that are complementary in principle: the page's text (an accessibility tree) is precise, and a screenshot shows what the text omits. A third option fuses them, overlaying numbered marks on the screenshot and listing the marked elements as text (Yang et al., 2023; Koh et al., 2024). Practitioners typically pick one of the three for a whole deployment. We ask the question this choice raises: *is the choice worth making per task, and can it be made?*

Answering it requires keeping apart three quantities that a single leaderboard number merges:

1. **Representation value**: does the screenshot change success, holding everything else fixed?
2. **Preference reliability**: when one representation beats another on a task, is that a stable property of the task, or a property of one run?
3. **Routing gain**: can a policy that chooses per task, using only what is known before it runs, beat the fixed choices?

A hindsight ceiling picks, for each task, whichever representation happened to succeed. However large, it answers none of the three questions, because it rewards run-to-run luck as much as structure. We therefore rerun identical conditions and judge each against what a rerun alone changes.

Findings. (i) No choice wins everywhere: the fused mode leads clearly on classifieds, while text leads on WebArena's reddit tasks only by margins whose intervals cross zero (§3). (ii) The screenshot has a specific value that can be isolated and predicted. On the classifieds tasks that a zero-cost intent rule flags, adding the screenshot to an unchanged text payload and prompt raises success by 25–30 points for the two strongest backbones (11 for a 4B model), against 5–10 on the other tasks. The effect reproduces on a full rerun for the 235B backbone and is larger for the stronger backbones we tested. On reddit the same rule finds nothing (§4). (iii) Per-task preferences between representations are reproducible beyond an additive difficulty model. Wherever all modes were rerun, we reject a no-interaction null conditioned on every task's success count and on every mode's success count in each run. Yet a single run measures these preferences with reliability 0.31–0.50, and a decision study projects that labels averaged over two to three runs would pass 0.5. Over all

six modes, single-run labels also flip most on exactly the tasks where the choice can change success. The additive difficulty model already predicts this concentration, so any router trained on single runs inherits it. In this sense routing is least learnable where it is most valuable (§5). (iv) Two lightweight routers that fix their operating point on training data show no reliable gain over the fixed representations and their random mixtures. In hindsight, only one pooled gain survives correction across curves (three modes, $p = 0.001$). It comes from a router that only decides which tasks deserve the expensive mode, holds under billed cost only, and is not captured without looking at test outcomes (§6).

Contributions. We measure text, screenshot and fused observation across three sites and four backbones, using reruns as controls and screenshot-free ablations to isolate the screenshot. We estimate the screenshot's value with a matched contrast and give an ex-ante rule that predicts it on one site. We test whether per-task representation preferences are reproducible, with a decision study of how many runs a reliable label needs. We compare deployable policies against fixed choices *and their random mixtures*, the comparison a routing claim has to win. We take this baseline from model-routing benchmarks and apply it to observation choice (§6), and distil these comparisons into a four-step protocol (§7). Findings (iii) and (iv) are the *bounds* of our title: limits on how reliably one run labels a per-task preference and on what routers trained on such labels achieve.

2 Setup

Sites and tasks. We use three sites of VisualWebArena (Koh et al., 2024): classifieds (224 scored tasks), reddit (203) and shopping (432). We add the reddit site of WebArena (Zhou et al., 2024) (104 tasks: the 106 minus the two unachievable tasks, whose reference answer is "N/A"). VWA-reddit and WA-reddit are different task sets on the same application, so they are not two independent benchmarks. Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration and, for one reddit task, over agent trajectories (Appendix A).

Backbones and cells. B0 = Qwen3-VL-235B (API-served), B1 = Qwen3-VL-4B and B2 = Gemma-3-4B (both served locally), and B5 = GPT-5.6 (API, classifieds only). A *cell* is one site × backbone pair. The eight core cells, each run in every observation mode, are VWA-classifieds and VWA-reddit × {B0, B1, B2}, plus WA-reddit × {B0, B1}. B5-classifieds and the two shopping cells extend them (§4, §6) and are never pooled with the eight.

Observation modes. The deployment question has three answers (Figure 1):**DOM** (the page's accessibility tree as text), **Vision** (the raw screenshot, with an empty text payload) and **SoM** (a set-of-marks screenshot plus the list of marked elements as text; Yang et al., 2023). Three further modes carry no screenshot and serve as controls; the one used throughout is **P-SoM**, which is SoM with the screenshot removed. Marks and prompt are the same, so SoM−P-SoM isolates the image (Appendix C lists all six). "Text" modes withhold the *page* screenshot only: a reference image that is part of the task ("find this item") is given in every mode.

One plain agent. One model call per step emits a thought, an action and its arguments; the prompt holds the instruction, the mode's system prompt, the last eight steps and the current observation. There is no planner, memory, reflection or retry policy, so conditions differ in the observation and in how actions reach the browser. Lacking element ids, Vision clicks by coordinates, and such clicks succeed less often in this harness than element-id clicks. Decoding is greedy, the viewport is 1280×720, and the text observation covers only the current viewport and is never truncated. The nominal budget is 30 steps, but a parser defect let some extra actions through, mostly for GPT-5.6 (Appendix E). Appendix B gives the full contract.

Reruns. We reran identical conditions, 27 registered same-condition pairs in all. In three cells (VWA-classifieds B0, VWA-reddit B0 and WA-reddit B1), every mode was run twice. A rerun uses the same tasks and code on freshly reset sites, so it exposes what one run of the agent does not determine. In our data a rerun changes between 0 and 14% of outcomes, depending on mode and cell (Appendix F). Sites are reset before every run; within a run only classifieds resets between tasks, so reddit and shopping carry state from earlier episodes (Appendix E).

Scoring and cost. The evaluators are binary. Cost is per episode, in US cents: the API charge for B0 and B5, and for B1 and B2 a token-priced

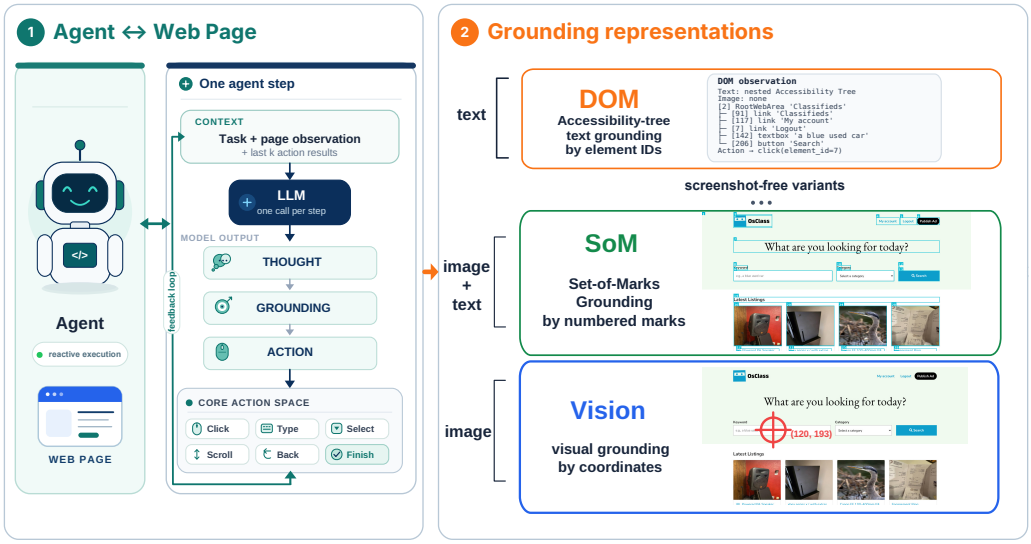


Figure 1: The agent and its observation modes. **(1)** One model call per step returns a thought and a grounded action; the prompt holds the task, the current observation and the last k=8 steps. **(2)** The three deployment modes ground an action by element id (DOM), by numbered mark (SoM) or by screen coordinates (Vision). The screenshot-free modes (P-SoM and two others; Appendix C) withhold the page screenshot.

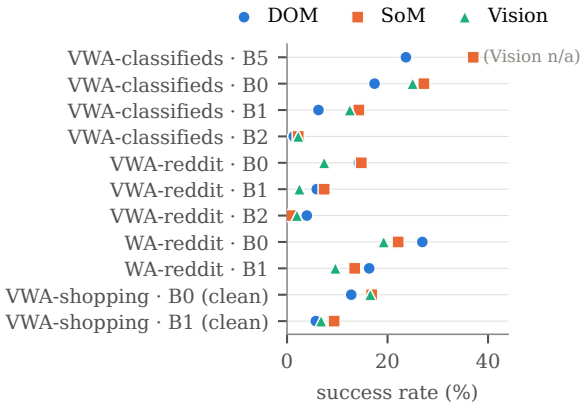


Figure 2: Success rate of the three deployment choices per cell. Shopping cells drop 46 state-sensitive tasks and every task the search-box defect touched in any mode (Appendix E). B5's Vision run is excluded for a coordinate-contract defect. Sources: `routing_three_arm`, `routing_extension_cells`.

proxy for local serving. The proxy charges hardware amortisation plus electricity per input and output token (per-cell values in Table 4). Costs are compared within a cell and never pooled across cells. §6 repeats the routing analysis under GPU-time and wall-clock cost.

3 No observation choice is best everywhere

The fused mode clearly leads text on classifieds (Figure 2). On reddit, text's leads all have intervals crossing zero once credit inherited from earlier episodes is removed. On classifieds the fused mode is the best of the three choices for B0 and B1, and beats DOM for GPT-5.6, whose Vision run is excluded. B2 solves too few tasks to order the modes. The margin over text is not small: SoM−DOM is +9.82 pp [+3.57, +16.07] for B0 and +8.04 pp [+3.57, +12.95] for B1 (paired task bootstrap). On WA-reddit the order flips and text leads (SoM−DOM −4.81 pp for B0 and −2.88 pp for B1, both intervals crossing zero). On VWA-reddit the weakest backbone does worse with the fused mode (−2.96 pp [−5.91, −0.49]). But three of its eight DOM successes were credited although their episodes never visited the target forum; an earlier episode had subscribed to it. Without them the gap is −1.48 pp [−3.45, +0.49] (Appendix E). On shopping the fused mode is best or tied with Vision, on the filtered and on the full task sets (Figure 2, Table 4).

Two cautions keep this descriptive. Cells on the same site share their tasks, so the eight are not eight independent replications. And there are four screenshot-free modes, not one. Comparing the best of four text variants with a single screenshot mode would introduce a selection effect, so we compare one mode per deployment choice, as practitioners do.

This section establishes only that the choice matters and depends on the workload. The rest of the paper asks whether the choice can be made per task.

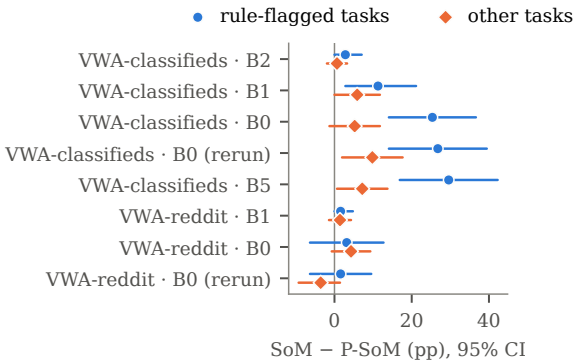


Figure 3: Value of the screenshot alone (SoM−P-SoM: same marks, same prompt) on the tasks a zero-cost intent rule flags, against the other tasks. Classifieds is out-of-sample for the rule. "Rerun" is a full repeat of both modes. VWA-reddit B2 is omitted: neither SoM nor P-SoM solves a flagged task. Source: `visual_intent_routing`.

4 The screenshot has a value that can be isolated and, on one site, predicted

A matched contrast. Neither contrast between the deployment choices of §3 isolates the image. SoM−DOM also changes the text payload and the prompt. Vision−DOM also changes the text and how clicks reach the page (by coordinates, since Vision has no element ids). SoM and P-SoM differ only in the annotated screenshot, so their difference is the value of the image alone. Chi et al. (2024) already compared screenshot-plus-text with text only on a subset of VisualWebArena tasks for one backbone (50.0% against 46.3%, without set-of-marks). We hold the set-of-marks text and prompt fixed, use four backbones and full task sets, rerun the contrast, and ask *where* the image pays off.

An ex-ante rule. A regular expression over the task intent flags tasks that ask about an image, a photo, a colour or a count of things shown, excluding tasks that already carry a reference image (Appendix D). It reads only the task configuration: no model call and no episode. We wrote it to diagnose reddit failures a week before we first used it to partition tasks, and never inspected its classifieds hits, so classifieds is out-of-sample for it.

Result. On the 71 classifieds tasks the rule flags, the screenshot raises success by +25.35 pp [+14.08, +36.62] for B0, against +5.23 pp on the other 153 tasks (Figure 3). The effect reproduces on a full rerun of both modes (+26.76 pp [+14.08, +39.44], from 23 against 4 successes). It is larger for the stronger backbones we tested: +2.82 pp for B2, +11.27 pp for B1, +25.35 pp for B0 and +29.58 pp [+16.90, +42.25] for GPT-5.6. The backbones differ in more than capability, so this is an association across the four we ran, not a scaling law. For GPT-5.6, a parser defect let some actions through beyond the step budget, in both modes alike (Appendix E). On VWA-reddit the same rule flags 63 tasks and separates nothing: +3.17 pp on flagged tasks against +4.29 pp elsewhere for B0, +1.59 pp against −3.57 pp on its rerun. On shopping the flagged tasks carry at most five successes per mode, too few to read, and on WA-reddit the rule flags only 5 of 104 tasks.

Reading. On classifieds we can predict, before anything runs, where the screenshot pays off, and it pays off more for the stronger backbones we tested. On reddit we cannot. The screenshot's value is therefore real and partly foreseeable, but it belongs to a kind of task on one site. The rule is also not a router. On classifieds, switching between DOM and Vision by this rule does not beat simply using SoM (for B0, 55 against 61 successes). Vision solves two more flagged tasks than SoM, but DOM solves eight fewer of the rest. The open question is whether *individual* tasks have stable preferences beyond their difficulty (§5).

5 Per-task preferences are reproducible, but one run measures them poorly

§4 found structure at the level of task *types*. Routing per task needs more. An individual task must prefer one representation over another beyond what an additive model of task difficulty predicts, and the preference must be stable enough to learn from.

A null without task-by-mode structure. In the three cells where every mode was run twice, let $Y_{imr} \in \{0, 1\}$ be the outcome of task i under mode m in run r. Under

$$\operatorname{logit} \Pr(Y_{imr} = 1) = a_i + b_{mr},$$

every task has its own difficulty and every (mode, run) column its own easiness, but there is no persistent task-by-mode interaction. This is the Rasch model, a special case of the multidimensional item-response models that item-response routers fit (Song et al., 2025). Here it is the null, and we test for structure beyond it. The row and column sums are sufficient for this model, so, given them, every 0/1 matrix with those sums is equally likely. We approximate this conditional null by curveball

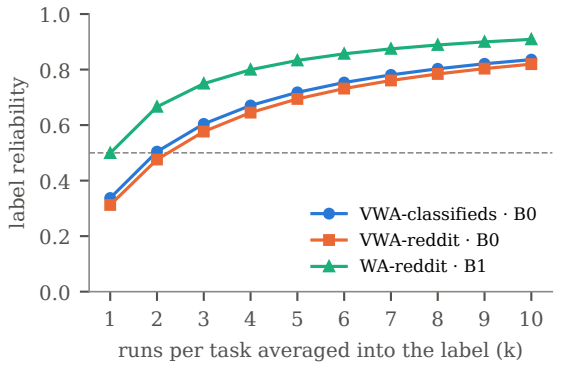


Figure 4: Decision study for the three deployment modes: reliability of a task-by-mode label averaged over k runs, $\sigma^2_{\text{int}}/(\sigma^2_{\text{int}} + \sigma^2_{\text{err}}/k)$. A single run gives 0.31–0.50. Source: `routing_three_arm`.

Markov-chain sampling (Strona et al., 2014), fitting nothing. The statistic is the run-to-run covariance of each task's mode profile, the reproducible part of the interaction: $\hat{\sigma}^2_{\text{int}} = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}})/2$ from the two-way task × mode ANOVA with two runs.

Result. Over the three deployment modes the interaction exceeds the null in all three cells (classifieds B0 $\hat{\sigma}^2_{\text{int}} = 0.033$ against a null 95th percentile of 0.015; VWA-reddit B0 0.018 against 0.009; WA-reddit B1 0.030 against 0.014; $p \leq 0.002$ each, Holm across cells). Over all six modes the same holds (Appendix F). Preferences between representations are therefore a reproducible property of tasks that an additive difficulty model does not capture. The test does not say what the extra structure is. A task-specific preference would reject this null, but so would a mode whose success responds more steeply to difficulty than another's. Only the first is a representation preference in the narrow sense. The preferences also mostly persist where they can be observed directly. Over all six modes, we pick for each task the cheapest mode that succeeded on one run and score it on the rerun. On VWA-reddit B0 and WA-reddit B1, this choice beats the frontier of the fixed modes and their random mixtures (§6) by 2.4–10.5 points. On VWA-classifieds B0 it lands +3.57 or −0.56 points relative to the frontier, depending on which run supplies the choice. That choice uses the task's own label, which no deployment has, so it measures persistence, not an attainable policy.

But a single run measures them poorly. As in a generalisability decision study (Brennan, 2001), the same ANOVA gives the reliability of a task's mode profile averaged over k runs (Figure 4). From one run it is 0.34, 0.31 and 0.50 in the three cells; the projection crosses 0.5 at two to three runs (bootstrap ranges 2–4, 2–7 and 1–3). Over six modes it is lower still, 0.14–0.46, with up to six runs projected. This is the reliability of the interaction profile, not of the mode a router picks. It therefore bounds how noisy single-run training labels are, but does not predict when a router starts to win.

Labels flip most where the choice can change success. A representation choice can change a run's success only on tasks that some modes solve and others do not; on the other tasks it can change only the cost. Over all six modes, the tasks split this way on the first run are 24–39% of each of the three fully rerun cells, but 78–84% of the tasks whose outcome changes in some mode on the rerun. Of the split tasks, 74–86% change, against 6.5–10.3% of the rest. This concentration needs no task-by-mode structure. A binary outcome varies most at intermediate difficulty, and the same no-interaction null over six modes concentrates the changes on these tasks as strongly. The observed ratio of the two change rates, 7.9–11.4, lies between the null's 5th and 95th percentiles in every cell. Still, these tasks show some task-by-mode structure: they change less often than under the null, below its 5th percentile in every cell.

The routing target thus exists. But single-run training labels measure it with a reliability of only 0.31–0.50, and they flip most where the choice can change success. In this sense, routing is least learnable where it is most valuable. This compares tasks within a cell, not cells or backbones, and concerns the labels any router learns from; we do not claim that it alone explains the results of §6.

6 Deployable routers show no reliable gain

The comparison a router must win. A router that sends some tasks to an expensive mode and others to a cheap one trades success for cost. Beating every fixed mode on both axes is too strict. Beating the fixed modes one at a time is too lenient, because sending a random share of tasks to each of two modes is also a deployable policy. We therefore compare against the *frontier of the fixed modes and all their random mixtures*: the upper concave envelope of the fixed modes' points in the (cost, success) plane. Model-routing benchmarks and method papers use this non-decreasing convex

hull as a baseline (Hu et al., 2024; Jitkrittum et al., 2025). We report how far a router rises above the frontier (Figure 5a; definitions in Appendix I).

Routers. Both routers use 18 features available before a task runs: 14 intent-keyword indicators, the intent length, and three statistics of the first page. *Per-arm heads* fit one logistic success model per mode and send a task to the cheapest mode whose predicted success clears a threshold. *Triage* predicts whether any mode can solve the task and sends predicted-hopeless tasks to the cheapest mode and the rest to the best one. All predictions are out-of-fold, over five folds of tasks. Both learn from one run's outcomes, which §5 found to measure per-task preferences poorly.

Room above the frontier exists in hindsight. Pooled over cells, only triage finds room that clears a null, and only under billed cost (see Robustness). Triage's decision is about difficulty, not representation: it decides which tasks are not worth the expensive mode. Sweeping each router's threshold traces a curve. Over the three deployment modes, the triage curve pooled over the eight cells rises 1.69 points above the frontier at low budget (Figure 5b). This rise clears a label-shuffle null that refits the whole curve ($p = 0.001$; Table 1). The per-arm heads reach 0.86 points ($p = 0.072$). That the per-arm heads miss their null does not show that representation fit carries nothing: a difference in significance is not a significant difference.

A deployable router does not capture it. A curve's best point is chosen after seeing test outcomes. When the operating point (threshold) is instead chosen on training folds only, the pooled gain falls to +0.23 points for per-arm heads and +0.59 for triage, against 0.86 and 1.69 in hindsight (Table 1). The per-arm interval has its upper end below its point estimate, a sign that the task bootstrap is biased when the threshold is re-selected in every draw. We therefore read all deployable intervals as magnitudes and make no significance claim from them, in either direction. Over all six modes both deployable routers sit slightly below the frontier (-0.38 and -0.39). The pattern holds for the strongest backbone: for GPT-5.6 on classifieds, where the screenshot's value was largest (§4), the deployable routers lose 1.63–2.45 points. On both shopping cells, filtered or on all tasks, they stay within 1.3 points of the frontier.

Robustness. Over six modes, holding out tasks of the same template together lowers the per-arm heads' mean deployable gain from -0.38 to -1.60 points. Fitting on one run and scoring on its rerun, with the same template folds, gives per-cell point estimates of both signs, which average below zero (Appendix G). Repricing cost changes the hindsight result but not the deployable one (Table 3). The pooled triage curve no longer clears its null under wall-clock cost (+0.68 points, $p = 0.42$) or under GPU time on the five locally served cells (+1.24, $p = 0.39$). The per-arm heads stay just short of significance under every basis ($p = 0.072$–0.077). Under every basis, the deployable gains pooled over all its cells stay between -0.5 and $+1.5$ points. Dropping the two cells of the weakest backbone (B2) leaves the billed-cost picture unchanged (triage curve +1.85, $p = 0.001$; deployable +0.50 and +0.39).

7 A protocol for multi-representation claims

We kept the three quantities of §1 apart by measuring each against its own control. We suggest four steps for any claim that choosing among several agent configurations (*arms*) per task would pay.

1. **Report reruns per arm and per cell**, not one flip rate for a whole study: in our data a rerun changes between 0 and 14% of outcomes depending on arm and cell (Appendix F).
2. **Isolate the component of interest with a matched ablation** (here SoM−P-SoM, which removes only the screenshot) before attributing a difference to it.
3. **Test for per-task structure against a null that conditions on difficulty**, and use a decision study to say how many runs per task make the label reliable. A router cannot reach a hindsight ceiling computed from one run.
4. **Choose the operating point on training data, and score the deployable policy against the fixed arms and their random mixtures**, never only against the best single arm or a hindsight curve.

Applied here, the protocol says that the screenshot raises success most on a predictable class of classifieds tasks. It also says that per-task representation preferences are reproducible but single-run labels measure them with a reliability of only

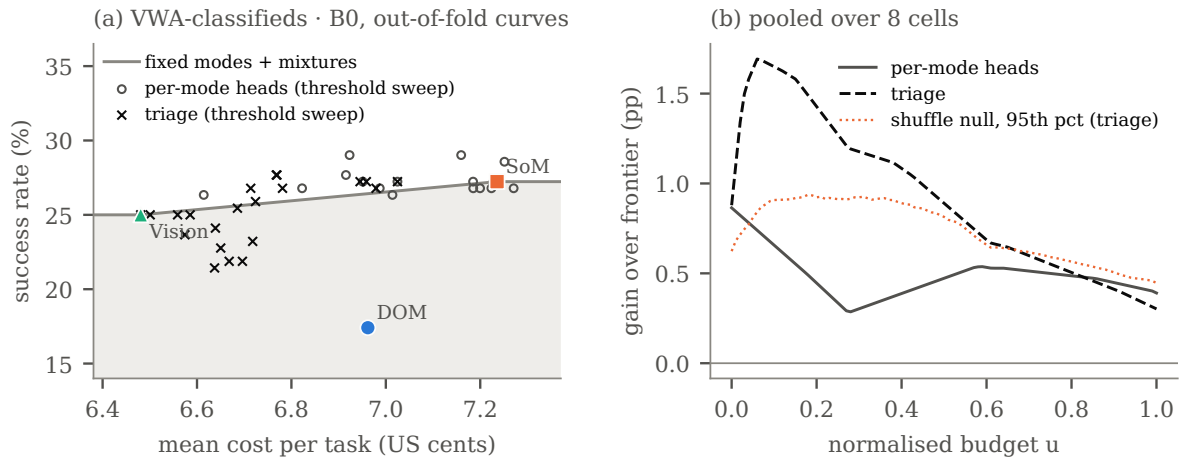


Figure 5: (a) One cell's cost–success plane: the three deployment modes, the frontier of the modes and their random mixtures (shaded: attainable without routing), and the out-of-fold points of the two routers as their threshold is swept. (b) Gain of the frontier when each router's curve is added, averaged over the eight cells, on a budget normalised from the cheapest mode ($u = 0$) to the dearest ($u = 1$). The null line is the pointwise 95th percentile of a label-shuffle null that refits the whole triage curve; the p-values in Table 1 test the maximum over u. Both panels choose the best point after seeing test outcomes; Table 1 gives the deployable version. Source: `routing_three_arm`.

0.31–0.50. Over six modes, those labels also flip most where the choice can change success, as the additive difficulty model predicts. In this sense routing is least learnable where it is most valuable. With single-run labels and pre-run features, the routers we tested do not reliably beat choosing one representation, or a fixed mixture, for the whole workload.

8 Related work

How web agents observe the page. Agents have read HTML or accessibility trees (Deng et al., 2023; Yang et al., 2025), screenshots grounded to elements (Zheng et al., 2024; He et al., 2024; Gou et al., 2025), or set-of-marks overlays that fuse the two (Yang et al., 2023; Koh et al., 2024). A parallel line shrinks or reshapes the text observation (Kerboua et al., 2025; Schiepanski and Piël, 2025; Enomoto et al., 2026), and some systems combine several grounding routes (Li et al., 2026a). OSWorld compares screenshot, accessibility-tree, combined and set-of-marks inputs across models and finds the best input model-dependent (Xie et al., 2024), and Chi et al. (2024) compare screenshot-plus-text with text only on a VisualWebArena subset. Closer to adaptivity, Li and Shi (2026) gate visual observation components during computer-use tasks, and Liu et al. (2026) route each action of a computer-use agent among vision-language models. Neither chooses between text and screenshot per task before it runs. The works above compare observation designs by benchmark score or adapt them online. OSWorld already shows that the best observation varies. We contribute a matched removal of the screenshot, repeated task-level outcomes, a test of task structure beyond difficulty, and a deployable per-task choice judged against mixtures of the fixed choices.

Routing and cascades. Choosing among models per query to trade quality for cost is well studied for language models (Chen et al., 2024; Ong et al., 2025; Ding et al., 2024; Gupta et al., 2024; Peale et al., 2026); see Moslem and Kelleher (2026) for a survey. For web agents, Li et al. (2025) learn query-specific routing for cost-sensitive agents, and Hajimiri et al. (2026) judge agent components at a matched budget. Early exit (Lu et al., 2025) stops agents that are unlikely to succeed, a decision about difficulty. In our data, such decisions are the only part of routing whose pooled gain over the frontier clears its null in hindsight. Routing benchmarks and method papers compare routers against random mixtures of fixed models (Hu et al., 2024; Jitkrittum et al., 2025). Under unified conditions, several learned routers fail to beat a simple baseline (Li et al., 2026b). Lee (2026) finds that assigning one model per task type in advance recovers most of the oracle routing gain on seven task types in three languages, and that reruns change the score of 5% of model–question pairs. Item-response routers esti-

cells	policy	curve max (p)	deployable [5%, 95%]
3 modes, 8 cells pooled	per-mode heads	+0.86 (0.072)	+0.23 [−2.20, +0.22]
	triage	+1.69 (0.001)	+0.59 [−1.53, +1.23]
6 modes, 8 cells pooled	per-mode heads	+0.67 (0.252)	−0.38 [−3.20, −0.14]
	triage	+1.24 (0.043)	−0.39 [−2.42, +0.81]
GPT-5.6, classifieds (5 modes)	per-mode heads	+0.45 (0.768)	−2.45 [−8.61, +4.24]
	triage	+3.86 (0.127)	−1.63 [−6.70, +5.74]
Shopping B0, clean (3 modes)	per-mode heads	+1.14 (0.062)	−0.18 [−2.45, +2.02]
	triage	+0.20 (0.496)	−0.87 [−2.63, +1.37]
Shopping B0, all tasks (3 modes)	per-mode heads	+1.09 (0.049)	+1.30 [−2.31, +1.85]
	triage	+0.18 (0.541)	−0.05 [−2.14, +1.45]
Shopping B1, clean (6 modes)	per-mode heads	+1.11 (0.143)	+0.56 [−4.69, +0.88]
	triage	+1.63 (0.158)	+1.19 [−4.10, +1.57]
Shopping B1, all tasks (6 modes)	per-mode heads	+0.48 (0.169)	−0.43 [−2.78, +0.73]
	triage	+0.69 (0.289)	+0.40 [−2.02, +2.02]

Table 1: Routing gain over the fixed modes and their random mixtures, in success-rate points. *Curve max*: the best point of an out-of-fold curve, chosen after seeing test outcomes, with its label-shuffle p (pooled rows: max over a normalised budget). *Deployable*: the operating point chosen on training folds only, task bootstrap. Sources: `routing_three_arm`, `representation_routing_frontier`, `routing_extension_cells`.

mate query difficulty and model ability (Song et al., 2025). We route over observation representations rather than models, take the mixture baseline from this literature, and add what it rarely has: reruns of identical agent conditions and a test of per-task structure beyond difficulty.

Reliability of agent evaluation. Repeated trials of the same agent disagree, which τ-bench measures with pass^k (Yao et al., 2025). Inference itself is not deterministic even at fixed settings (He and Thinking Machines Lab, 2025; Yuan et al., 2025), prompts change outcomes in ways unrelated to content (Sclar et al., 2024), and agent benchmarks carry evaluator and task defects (El Hattami et al., 2025; Bhat et al., 2026). Our reruns measure run-to-run disagreement for web agents at the task level and show its effect on per-task routing labels. Appendix E lists the harness defects we found.

Limitations

Coverage of reruns. The preference test and decision study (§5) rest on the only three cells with every mode run twice, two of which share a backbone. Of the remaining eight of the eleven cells, five have some modes rerun and three have none. On classifieds, B2 solves about 2% of tasks, so its two reruns cannot bound an effect.

Breadth. VWA-reddit and WA-reddit are task sets on the same application, so the study covers three applications, not four benchmarks. GPT-5.6 was run on classifieds only, and its Vision run is excluded for a coordinate-contract defect. A parser defect also let GPT-5.6 take actions beyond the 30-step budget in every mode (Appendix E). The ex-ante rule of §4 works on classifieds only.

Shopping filtering. The clean shopping cells drop every task on which the search-box defect fired in any mode, as well as the 42 grid-order-sensitive and 4 wishlist tasks (Appendix E). Whether the defect fired depends on what the agent typed, so the filter selects tasks by behaviour under the conditions being compared. The clean cells estimate the comparison on an observed defect-free subset, not on the benchmark, and B0 and B1 keep different subsets (368 and 192 tasks). Results on all tasks are reported alongside (Tables 1 and 4); no conclusion rests on the filtered cells alone.

What the routing result does not show. The routers use 18 pre-run features and logistic models; richer features, language-model routers or multiple runs per training task may do better. With single-run labels and these features, the routers do not reliably beat the fixed choices and their mixtures. The result does not show that per-task routing is impossible. Step-level switching is out of scope.

Inherited state. Reddit and shopping are reset before each run but not between tasks, so an episode can be credited for state an earlier episode created. A mechanical audit removes reddit successes credited without visiting the target forum and changes one comparison in the paper (§3). The audit cannot see an episode that visits an already-subscribed forum and leaves without acting, so red-

dit outcomes may still carry inherited credit, and reruns may reproduce it.

Exploratory analyses. The preference test, the reading of label changes in §5, the frontier comparisons and the deployable-router estimates were designed after the runs were collected. Only the preference test is corrected across cells (Holm). The frontier p-values account for choosing each curve's best point but are not corrected across curves. Only the pooled three-mode triage curve survives Holm correction across the 14 curve maxima of Table 1. The bootstrap intervals for policies with a selected threshold are read as magnitudes, not tests.

Cost. API charges and local-serving estimates are different quantities and are never pooled. Local cost also depends on the pricing basis: the pooled three-mode triage curve clears its null under billed cost but not under wall-clock time or under GPU time on the locally served cells (§6).

Ethics statement

All episodes ran against self-hosted copies of the VisualWebArena and WebArena sites, and no human subjects were involved. No real account or person was contacted, but agents could follow external links off the self-hosted sites. In the audited runs, 3–6% of reddit episodes and at most 0.3% of classifieds episodes visited live public pages. Sites were reset before each run. The API-served backbones were used under their providers' terms. Agents that act on websites can cause harm when deployed against live services; this study measures observation choices on self-hosted sites and releases no agent that acts outside them.

Availability. The anonymous supplementary material contains the agent and run configurations, per-episode outcomes and costs for every run used, the analysis scripts, and every analysis product cited in the text. A guide in the supplement maps each number in the paper to the product field it comes from. One script regenerates the figures and tables from the products. Full trajectories (screenshots and page text) are too large for the supplement and will be released with the camera-ready version.

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A Task universes and exclusions

Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration or, for one reddit task, over agent trajectories: classifieds 224 of 224; VWA-reddit 203 of 205 (one task whose evaluator never checks the requested action, and one cross-site task answerable from memory); shopping 432 of 435. For every site the collected tasks are the benchmark's tasks minus its unachievable tasks (reference answer "N/A"), which we drop because our agent prompt does not offer N/A as an answer (WA-reddit: 104 of 106). Every analysis records the task set it was built on and fails if denominators differ.

B Agent and run contract

Every condition uses the same agent loop and settings; only the observation mode and backbone vary, apart from one output-format line of the system prompt (B0 is told to answer through a tool call, GPT-5.6 to emit exactly one JSON object). Decoding is greedy (temperature 0, at most 4096 new tokens per step). The browser viewport is 1280×720, and the page is observed 0.5 s after each action. The prompt holds the task instruction, the mode's system prompt, the last eight steps (action, outcome and resulting URL) and the current observation; the text observation (accessibility tree or marked-element list) covers only the current viewport and is never truncated. The nominal step budget is 30; three consecutive unparsable outputs end an episode. B1 (Qwen3-VL-4B-Instruct) and B2 (Gemma-3-4B-it) run locally from pinned model revisions; B0 (Qwen3-VL-235B-A22B) and B5 (the mid-price GPT-5.6 tier) are reached through a hosted API under a fixed model identifier. Each run uses seed 42, starts from freshly reset sites, and records the code version, configuration and environment with every episode. The exact prompts and configuration files are in the supplementary material.

C The six observation modes

Four modes carry no page screenshot and cross two factors: the text payload (the accessibility tree, or the flattened list of set-of-marks elements) and the prompt family (DOM or SoM). DOM is tree + DOM prompt; P-prompt is tree + SoM prompt; P-text is marks + DOM prompt; P-SoM is marks + SoM prompt. SoM adds the annotated screenshot to P-SoM; Vision is the raw screenshot with an empty text payload. A task's own reference image, when it has one, is given in every mode.

D The intent rule

A task is flagged if its intent matches

```
\b(image|picture|photo|screenshot)\b|
\bcolou?r of\b|
\bhow many\b[^.]{0,40}\bin (?:the|this)\b
```

(case-insensitive) and the task carries no reference image. It flags 71 of 224 classifieds tasks, 63 of 203 VWA-reddit tasks, 56 of 432 shopping tasks and 5 of 104 WA-reddit tasks. It was written while auditing reddit failures; classifieds is out-of-sample.

E Harness defects and how cells were filtered

- **GPT-5.6 Vision**: the run used a broken coordinate contract and is excluded.
- **GPT-5.6 step budget**: when the model emitted several valid actions, the parser executed one but booked the step as a free wait. Summed over a whole condition, GPT-5.6 had 49–89 such steps (1.2–2.4% of its steps; most of them executed an action, and some episodes ran 31–32 steps in all), B2 had 2–24, and B0 and B1 none (counts from the classifieds runs). It affects SoM and P-SoM alike.

- **Reddit state**: subscriptions persist across the episodes of a run. Of the reddit successes scored by the subscribed-forum list, six were credited although their episodes never visited the target forum, three of them for B2 with DOM; §3 reports the B2 comparison with and without them. The audit cannot detect an episode that reaches an already-subscribed forum and leaves without acting (one such case is confirmed on WA-reddit).

- **Shopping search box**: typing into a search box that already holds a query submitted old + new text ("chair" + "chair with wheels" → "chairchair with wheels") in 43% of such submissions, far more often in text modes than in Vision. A detector reproducing the defect's per-mode counts marks affected episodes; the *clean* shopping cells drop every task affected in any mode (18 tasks for B0, 201 for B1).

- **Shopping state**: the category grid order differs between runs (42 order-sensitive tasks dropped), and the wishlist is not reset between tasks (4 tasks dropped); with all three filters, B0 keeps 368 of 432 tasks and B1 192.

F Six-mode results

Table 2 gives the success rate of every mode in every cell. Over all six modes, the no-interaction null of §5 is rejected in all three fully rerun cells ($p = 0.0005$, 0.016, 0.0005; Holm), with single-run reliability 0.29, 0.14 and 0.46 and three, six and two runs per task needed to reach 0.5. Across the 27 registered rerun pairs, a rerun changes between 0 and 14% of task outcomes, depending on arm and cell.

G Template and rerun robustness

Averaged over cells, the deployable excess over the frontier with folds grouped by task template is -1.60 (per-arm heads) and -0.84 (triage) points over six modes, and -0.64 and $+0.29$ over three. Fitting on one run and scoring on its rerun, with the same template folds and averaged over the three fully rerun cells, gives -0.90 and -1.60 over six modes and -0.60 and -0.56 over three. Over six modes, every per-cell template-bootstrap interval (90%) includes zero; the three-mode readings are point estimates.

H Per-cell results

Table 4 lists the success rate and cost behind Figures 2 and 5. In shopping the fused mode is cheaper than text: its marked-element list is shorter than the accessibility tree of a product-dense page.

I Estimators

Frontier. For a cell with arms m of mean cost c_m and success rate s_m, $F(c)$ is the upper concave envelope of $\{(c_m, s_m)\}$, cut at the highest-success arm (constant beyond it) and equal to the cheapest arm's success below the cheapest cost. Between the cheapest and the highest-success arm, every point of F is reached by sending a random share of tasks to each of two arms; no mixture costs less than the cheapest arm, so F below that cost is a convention that favours the fixed arms. A policy with mean cost c and success s has excess $s - F(c)$.

Router curves. Features are standardised within each training fold and fed to an L2-regularised logistic regression ($C = 1$); predictions are out-of-fold over five folds of distinct tasks. *Per-arm heads*: for each threshold $\tau \in \{0, 0.05, \ldots, 1\}$ a task goes to the cheapest arm (by training-fold mean cost) whose predicted success is at least τ, and to the training fold's best arm if none is. *Triage*: one model predicts whether any arm solves the task; for each quantile $q \in \{0, 0.05, \ldots, 1\}$ of its score, tasks below the quantile go to the cheapest arm and the rest to the best arm.

Pooled gain. Within a cell, cost is normalised as $u = (c - c_{\min})/(c_{\max} - c_{\min})$ over the fixed arms, which makes cells with different cost units comparable without pooling their costs. The gain at u is the envelope of the fixed arms together with the curve's points minus the envelope of the fixed arms alone, on 101 grid points; the pooled curve is the equal-weight mean over cells and its statistic is the maximum over u. The null permutes each task's outcomes and costs against its features and refits the whole curve ($B = 1000$; $p = (k+1)/(B+1)$; for the pooled statistic each null draw is averaged over cells before the maximum).

Deployable policy. In each outer fold the threshold (or quantile) is the one with the largest excess over the *training* tasks' own frontier, using in-sample predictions on those tasks; it is then applied to the held-out fold. The five folds' decisions form one policy over all tasks, scored against the

		no screenshot				screenshot	
cell	n	DOM	P-prompt	P-text	P-SoM	SoM	Vision
cls B0	224	17.4†	19.6†	15.6†	15.6†	27.2†	25.0†
cls B1	224	6.2†	6.7	7.6	6.7	14.3†	12.5†
cls B2	224	1.3	1.8	0.4	0.9	2.2†	2.2†
VWA-red B0	203	14.3†	12.3†	13.3†	10.8†	14.8†	7.4†
VWA-red B1	203	5.9†	5.4	5.9	5.9	7.4†	2.5
VWA-red B2	203	3.9	0.0	2.0	0.5	1.0	2.0
WA-red B0	104	26.9	26.0	35.6	25.0	22.1	19.2
WA-red B1	104	16.3†	16.3†	16.3†	11.5†	13.5†	9.6†
cls B5	224	23.7†	21.9	24.1	22.8	37.1	–
shop B0 clean	368	12.8	–	–	–	16.8	16.6
shop B0 all	432	11.3	–	–	–	14.8	14.4
shop B1 clean	192	5.7	7.3	6.8	5.2	9.4	6.8
shop B1 all	432	4.9	5.3	6.7	4.6	7.6†	5.6

Table 2: Success rate (%) of all six observation modes per cell (Appendix C); † marks a mode with a registered same-condition rerun (Appendix F). Shopping B0 was not run with P-prompt, P-text or P-SoM; GPT-5.6 Vision is excluded; shopping rows as in Table 4. Sources: `representation_routing_frontier`, `routing_extension_cells`, `noise_floor_inventory`.

		curve max (p)		deployable	
cost	cells	per-mode	triage	per-mode	triage
billed	all 8	+0.86 (.072)	+1.69 (.001)	+0.23	+0.59
	no B2 6	+1.15 (.071)	+1.85 (.001)	+0.50	+0.39
wall-clock	all 8	+1.28 (.077)	+0.68 (.418)	+0.10	−0.48
	no B2 6	+1.70 (.076)	+0.60 (.619)	+0.18	−0.95
GPU time	all 5	+1.61 (.076)	+1.24 (.387)	+1.47	+0.05
	no B2 3	+2.68 (.076)	+2.06 (.376)	+2.58	−0.56

Table 3: Three-mode routing pooled over cells under three cost bases, with and without the weakest backbone (B2). GPU time exists only for the locally served B1 and B2. Curve max: best point after seeing test outcomes, with its label-shuffle p; deployable: operating point chosen on training folds. Sources: `routing_three_arm` and its `_wallclock`, `_gpu_time` siblings.

full-sample frontier. Intervals resample tasks and rerun the whole pipeline ($B = 1000$, 5th and 95th percentiles); pooled intervals average the cells draw by draw.

No-interaction null and decision study. For the task × mode × run array with $r = 2$ runs, the two-way random-effects ANOVA gives $\hat{\sigma}^2_{\text{err}} = \text{MS}_{\text{err}}$ and $\hat{\sigma}^2_{\text{int}} = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}})/r$. The null matrices are drawn with curveball trades over the task-by-(mode, run) 0/1 matrix, which preserve every row and column total (four chains, each with 50 trades per row of burn-in and 5 per row between kept draws, 2000 draws in all; for the six-mode analysis a second sampler with four-fold thinning checks mixing). The reliability of a label averaged over k runs is $\hat{\sigma}^2_{\text{int}}/(\hat{\sigma}^2_{\text{int}} + \hat{\sigma}^2_{\text{err}}/k)$, and the number of runs needed to reach 0.5 is $\lceil \hat{\sigma}^2_{\text{err}}/\hat{\sigma}^2_{\text{int}} \rceil$.

		DOM		SoM		Vision	
cell	n	SR	cost	SR	cost	SR	cost
cls B0	224	17.4	6.96	27.2	7.24	25.0	6.48
cls B1	224	6.2	5.95	14.3	6.03	12.5	4.32
cls B2	224	1.3	7.68	2.2	9.08	2.2	7.07
VWA-red B0	203	14.3	10.15	14.8	11.05	7.4	9.81
VWA-red B1	203	5.9	7.33	7.4	8.00	2.5	5.24
VWA-red B2	203	3.9	9.48	1.0	11.16	2.0	6.83
WA-red B0	104	26.9	7.53	22.1	9.11	19.2	8.64
WA-red B1	104	16.3	6.58	13.5	7.94	9.6	4.47
cls B5	224	23.7	13.74	37.1	16.71	–	–
shop B0 clean	368	12.8	12.25	16.8	9.87	16.6	7.75
shop B0 all	432	11.3	11.99	14.8	9.77	14.4	7.48
shop B1 clean	192	5.7	10.60	9.4	7.75	6.8	4.69
shop B1 all	432	4.9	11.11	7.6	8.27	5.6	4.94

Table 4: Success rate (%) and mean billed cost per task (US cents) of the three deployment modes. B0 and GPT-5.6 (B5) costs are API charges computed from token counts and per-token prices; B1 and B2 costs are token-priced estimates for local serving, comparable within a cell only. Shopping: clean = tasks on which the search-box defect fired in no mode, also excluding the 42 grid-order-sensitive and 4 wishlist tasks (Appendix E), all = every scored task; GPT-5.6 Vision is excluded. Sources: `routing_three_arm`, `routing_extension_cells`.